

Better Deterministic Approximation for Meta-Submodular Function Maximization

Abstract

We study the submodular maximization problem with a family of utility functions. Meta-submodular functions are proposed in [14] to summarize the properties of the density functions. Before such functions are defined, authors in [13] use the one-sided smoothness to solve the metric diversity problems. And their randomized algorithm is proposed with constant approximation factor through Jump-Start double greedy. Furthermore, a deterministic algorithm is provided with the approximation ratio that is controlled by parameter γ . We extend the basic greedy framework which is insensitive to this parameter.

In this paper, we introduce the meta-submodular function maximization under matroid constraint. Although a competitive approximation ratio of $\Omega(1/\sigma^{3/2})$ is achieved in [13], their applications are limited to the monotone case. More generally, we analyze the maximization of non-monotone submodular function. We design a deterministic greedy algorithm by solving the linear program, and it has the better approximation ratio of $(1 + e^{-2})/4$ than $\gamma^2 2^{4\gamma}$ in previous work [14]. Our derandomization strategy guarantees $\mathcal{O}(n^2 k^2)$ queries during all the iterations. Therefore, the algorithm complexity depends on the time to solve each linear program.

Keywords and phrases Meta-Submodular Function; Deterministic Greedy; Linear Program

1 Introduction

A set function $f : 2^{\mathcal{V}} \rightarrow \mathbb{R}_+$ is submodular if $f(A) + f(B) \geq f(A \cup B) + f(A \cap B)$ for every pair of subsets $A, B \subseteq \mathcal{V}$. In general, we only consider the normalized submodular functions [16], i.e., $f(\emptyset) = 0$. For any element $v \in \mathcal{V}$, the marginal gain $f(v|S) = f(S \cup \{v\}) - f(S)$ of the arbitrary set $S \subseteq \mathcal{V}$ will determine the monotonicity of a submodular function. In other words, submodular function $f(\cdot)$ is monotone if $f(v|S) \geq 0$, otherwise non-monotone. A value oracle for $f(\cdot)$ allows us to query the value of $f(S)$ for any set S in constant time.

Beyond the simple marginal gain, the authors in [14] define the first-order difference (also called marginal gain or discrete partial derivative) of element v with respect to set S as:

$$F(v|S) := f(S \cup \{v\}) - f(S \setminus \{v\}) \quad (1)$$

Obviously, this approach is better at measuring the increased utility when in a continuous element space. Further more, they define the second-order difference of elements u and v with respect to set S as:

$$\begin{aligned} F(u; v|S) &:= F(u|S \cup \{v\}) - F(u|S \setminus \{v\}) \\ &= f(S \cup \{v\} \cup \{u\}) - f(S \cup \{v\} \setminus \{u\}) - f(S \setminus \{v\} \cup \{u\}) + f(S \setminus \{v\} \setminus \{u\}) \end{aligned} \quad (2)$$

Naturally, the meta-submodular function is defined according to both first-order difference and second-order difference. For $\gamma \geq 0$, a set function $f(\cdot)$ is γ -meta-submodular for all nonempty sets $S \subseteq \mathcal{V}$, when

$$F(u; v|S) \leq \gamma \cdot \frac{F(v|S) + F(u|S)}{|S|} \quad (3)$$

The parameter γ controls the density metric of the set function with size sensitive. Therefore, we first study the submodularity ratio of meta-submodular function to ensure that the greedy algorithm can obtain a constant approximation. In addition, the authors in [13] define a similar framework using the continuous smooth function, which is guided by the one-sided smooth parameter σ .

1.1 Related Work

When the submodular function $f(\cdot)$ is non-monotone, the best-known deterministic approximation ratio under the matroid constraint is $1/4$, proposed in [17], but with a high time complexity of $\mathcal{O}((n^4 \log n)/\epsilon)$. Recently, there appear studies aiming at designing more efficient and practical algorithms for this problem. In this line of studies, the elegant work of Mirzasoleiman et al. [18] and Feldman et al. [9] proposes the best deterministic ratio of $1/6$ and the fastest implementation with an expected ratio of $1/4$, and their algorithms can also handle more general constraints such as a p -set system constraint [15]. As regards the efficient randomized algorithms for the optimization problem, the algorithms in [4] also achieve a $1/4$ expected ratio with slightly worse time complexity of $\mathcal{O}(nr \log n)$. Feldman et al. [10] proposed a $1/(3 + 2\sqrt{2})$ expected ratio under the streaming setting. However, it remains an open problem to find the approximation ratio of this de-randomized algorithm for this problem with a non-monotone objective function.

Meta-submodular maximization problem is defined by Mehrdad, Richard and Bruce [14] for the first time. However, they only give a simple local greedy algorithm. Furthermore, they do not consider the more general form of matroid constraint in non-monotone case. Until better unconstrained submodular maximization algorithms come along [11, 3], submodular Welfare and submodular max-sat are proven to have $3/4$ approximation. Based on the solution of the linear program in [2], Nicolas and Allan [20] implements the de-randomized residual greedy. Furthermore, it appears that the approximation result [11] for the k partition matroid has $1 - (1 - 1/k)^k$. For the non-monotone case, Ganz et al. [12] have implemented the 0.27493 approximation ratio in the recent. The disadvantage is that researches on the non-monotone case are limited to randomized greedy algorithms.

Under the one-sided smoothness framework, Ghadiri, Santiago and Shepherd [13] study the maximization of a σ -semi-metric diversity function subject to matroid constraint. The disadvantage is that they are limited to monotone continuous cases, and a complex Swap Rounding [6] algorithm is applied to the final solution from fractional solution. Next, a parallel algorithm (Stochastic Parallel Frank Wolfe [24]) is designed for the non-monotone deterministic setting with approximation ratio $e^{-1} - e^{-2} - \mathcal{O}(\epsilon)$. Beyond the offline version, a similar algorithm is studied with online one-sided smooth function [23], which obtained a approximation factor closed to the bound of general submodular function. By adjusting the iterative step of the Jump-Start Frank Wolfe algorithm, a same approximation can be obtained in the case of a larger regret bound [1, 19]. Besides, the authors in [22] provides a good tool to maximize the multilinear extension of non-monotone set functions under one-sided smoothness.

1.2 Our Result

The main contribution of this work is showing that an easy-to-implement and fast deterministic greedy algorithm (i.e., Algorithm 1) has provable theoretical guarantees for the meta-submodular maximization problem. To the best of our knowledge, the following theorem shows a better result in the non-monotone setting with rank k matroid. Our algorithm is obtained by derandomizing the random greedy algorithm in [8], using the technique from [4].

► **Theorem 1.** *There is a deterministic algorithm achieves the $(1 + e^{-2})/4 - \mathcal{O}(1)$ approximation ratio for non-monotone meta-submodular maximization problem over the matroid. And the query complexity of the algorithm iterations is $\mathcal{O}(n^2 k^2)$.*

2 Preliminaries

In this paper, we focus on the optimization of non-monotone meta-submodular function, and the function $f(\cdot)$ could be monotone. For simplicity, let $S + v = S \cup \{v\}$, $S - v = S \setminus \{v\}$ and $\mathcal{V} = \{v_1, \dots, v_n\}$. Our intention is to model the meta-submodular maximization problem under a matroid, and the constrained optimization has the form $\max\{f(S) | S \in \mathcal{I}, |S| \leq k\}$. There are two properties about matroid:

► **Definition 2** (Matroid[5]). *The matroid $\mathcal{M} = (\mathcal{N}; \mathcal{I})$ is defined by a family of independent sets $\mathcal{I} \subseteq 2^{\mathcal{N}}$, satisfying:*

- *If $A \subseteq B$ and $B \in \mathcal{I}$, then $A \in \mathcal{I}$.*
- *For all $A, B \in \mathcal{I}$ with $|A| < |B|$, there exists an item $v \in B \setminus A$ such that $A + v \in \mathcal{I}$.*
- *The maximal independent sets of $\mathcal{I}$ are called bases of $\mathcal{M}$, and the last condition implies that they all have the same size, called the rank of $\mathcal{M}$, which we typically denote by k .*

► **Lemma 3** ([21]). *Let $\mathcal{M} = (\mathcal{N}; \mathcal{I})$ be a matroid and A, B be two bases of $\mathcal{M}$. Then there exists a bijective mapping $g : A \setminus B \rightarrow B \setminus A$ such that $A - v + g(v) \in \mathcal{I}$ for any $v \in A \setminus B$.*

For the non-submodular functions, submodularity ratio measures to what extent they have submodular properties., defined as:

► **Definition 4** (Submodularity ratio [7]). *The submodularity ratio of a non-negative set function $f(\cdot)$ is the largest scalar λ s.t.*

$$\sum_{v \in B} f(v|A) \geq \lambda f(B|A), \quad \forall A, B \subseteq \mathcal{V} \quad (4)$$

For a non-decreasing set function $f(\cdot)$, it holds: $\lambda \geq 0$; $f(\cdot)$ is submodular iff $\lambda = 1$.

3 Meta-Submodular Greedy Algorithm

3.1 Sensitive Submodularity Ratio

Let $f : 2^{\mathcal{V}} \rightarrow \mathbb{R}_+$ be a meta-submodular function. For any set $S \subseteq \mathcal{V}$ and elements $u, v \in \mathcal{V} \setminus S$; combining the Equation 2 and Equation 3, we have

$$\begin{aligned} f(S + v + u) - f(S + v) - f(S + u) + f(S) &\leq \frac{\gamma}{|S|} \cdot [f(S + v) + f(S + u) - 2f(S)] \\ f(S + v + u) - (1 + \frac{\gamma}{|S|}) \cdot f(S + v) &\leq (1 + \frac{\gamma}{|S|}) \cdot f(S + u) - (1 + \frac{2\gamma}{|S|}) \cdot f(S) \\ f(u|S + v) - \frac{\gamma}{|S|} \cdot f(S + v) &\leq (1 + \frac{\gamma}{|S|}) \cdot f(u|S) - \frac{\gamma}{|S|} \cdot f(S) \\ f(u|S + v) &\leq (1 + \frac{\gamma}{|S|}) \cdot f(u|S) + \frac{\gamma}{|S|} \cdot f(v|S) \end{aligned} \quad (5)$$

Then for two disjoint sets S and T with the same size k , where $S \cap T = \emptyset$ and $T = \{v_1, \dots, v_k\}$. Let $m := 2k - 1$, we successively apply Equation 5 to $f(v_k|S \cup \{v_1, \dots, v_{k-1}\})$. Each application of the equation generates two terms so the whole applications can be seen as a binary tree. For the nodes in the i -th level, we apply Equation 5 to remove v_i from the base. We get

$$f(v_k|S \cup \{v_1, \dots, v_{k-1}\}) \leq p^k \cdot f(v_k|S) + \sum_{i=1}^{k-1} p_i^k \cdot f(v_i|S) \quad (6)$$

125

126 where

$$\begin{aligned}
127 \quad p^k &= \prod_{i=1}^{k-1} \left(1 + \frac{\gamma}{m-i}\right) \\
p_1^k &= \frac{\gamma}{m-1} \prod_{i=2}^{k-1} \left(1 + \frac{\gamma}{m-i}\right) = \left(1 + \frac{2\gamma}{m-1} - 1 - \frac{\gamma}{m-1}\right) \prod_{i=2}^{k-1} \left(1 + \frac{\gamma}{m-i}\right) \\
p_2^k &= \left(1 + \frac{2\gamma}{m-1}\right) \frac{\gamma}{m-2} \prod_{i=3}^{k-1} \left(1 + \frac{\gamma}{m-i}\right) \\
&= \left(1 + \frac{2\gamma}{m-1}\right) \left(1 + \frac{2\gamma}{m-2} - 1 - \frac{\gamma}{m-2}\right) \prod_{i=3}^{k-1} \left(1 + \frac{\gamma}{m-i}\right) \\
&\vdots \\
128 \quad p_{k-1}^k &= \prod_{i=1}^{k-2} \left(1 + \frac{2\gamma}{m-i}\right) \frac{\gamma}{k} = \prod_{i=1}^{k-2} \left(1 + \frac{2\gamma}{m-i}\right) \left(1 + \frac{2\gamma}{k} - 1 - \frac{\gamma}{k}\right) \quad (7)
\end{aligned}$$

129 Besides that, we have

$$130 \quad p^k + \sum_{i=1}^{k-1} p_i^k = \prod_{i=1}^{k-1} \left(1 + \frac{2\gamma}{m-i}\right) \quad (8)$$

131 Due to $1+x \leq e^x$ and $\sum_{j=1}^n \frac{1}{j} \leq \ln(n) + 1$, we obtain

$$\begin{aligned}
132 \quad p^k + \sum_{i=1}^{k-1} p_i^k &\leq \prod_{i=1}^{k-1} \exp\left(\frac{2\gamma}{m-i}\right) = \exp\left(2\gamma \cdot \sum_{i=1}^{k-1} \frac{1}{m-i}\right) = \exp\left(2\gamma \cdot \left[\sum_{j=1}^{m-1} \frac{1}{j} - \sum_{j=1}^{m-k} \frac{1}{j}\right]\right) \\
133 \quad &\leq \exp(2\gamma \cdot [\ln(m-1) - \ln(m-k)]) = \left(\frac{m-1}{m-k}\right)^{2\gamma} = 4^\gamma \quad (9)
\end{aligned}$$

134 **► Lemma 5.** A non-negative meta-submodular function $f : 2^V \rightarrow \mathbb{R}_+$ has the sensitive submodularity ratio $\lambda \geq 4^{-\gamma}$.

135 **Proof.** By the Equation 6, we have

$$\begin{aligned}
136 \quad f(T|S) &= f(\{v_1, \dots, v_k\}|S) = \sum_{i=1}^k f(v_i|S \cup \{v_1, \dots, v_{i-1}\}) \\
137 \quad &\leq \sum_{i=1}^k [p^i \cdot f(v_i|S) + \sum_{j=1}^{i-1} p_j^i \cdot f(v_j|S)] \quad (10)
\end{aligned}$$

138 Note that the ordering of the elements $v_1, \dots, v_i$ is arbitrary. According to Equation 9, summing up the i inequalities obtained by rotating the ordering and dividing it by i , we get

$$\begin{aligned}
139 \quad f(T|S) &\leq \sum_{i=1}^k (p^i + \sum_{j=1}^{i-1} p_j^i) \frac{f(v_i|S)}{i} \leq \sum_{i=1}^k \frac{4^\gamma}{i} f(v_i|S) \\
140 \quad &\leq 4^\gamma \sum_{i=1}^k \left(\frac{1}{k} \sum_{i=1}^k \frac{1}{i}\right) f(v_i|S) \leq 4^\gamma \cdot \frac{\ln(k) + 1}{k} \sum_{i=1}^k f(v_i|S) \leq 4^\gamma \sum_{i=1}^k f(v_i|S) \quad (11)
\end{aligned}$$

141 ◀

■ **Algorithm 1** Modified Greedy

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- 1: **Input** $f(\cdot), \mathcal{M} = (\mathcal{N}; \mathcal{I})$.
 - 2: Initialize $\mathcal{D}_0 = \{(1, S)\}$, where S is a base containing only k dummy elements.
 - 3: **for** $i = 1$ to k **do**
 - 4: Let $M_i \subseteq \mathcal{N}$ be a base of $\mathcal{M}$ maximizing $\sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}}[f(v | S)]$.
 - 5: Construct mapping $g_{i,S}$ for every $S \in \text{supp}(\mathcal{D}_{i-1})$, where $g_{i,S}$ denotes the mapping defined in Lemma 3 by plugging $A = M_i$ and $B = S$.
 - 6: Find an extreme point solution of the following linear program:

$$\begin{aligned}
 & \max \sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}}[x(v, S) \cdot f(v | S)] \\
 & \sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}}[x(v, S) \cdot f(v | S)] \geq \frac{1}{k} \sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}}\left[\frac{4^\gamma}{i} \cdot f(v | S)\right] \\
 & \sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}}[x(v, S) \cdot f(g_{i,S}(v) | S - g_{i,S}(v))] \leq \frac{1}{k} \sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}}\left[\frac{4^\gamma}{i} \cdot f(g_{i,S}(v) | S - g_{i,S}(v))\right] \\
 & \mathbb{E}_{S \sim \mathcal{D}_{i-1}}[x(v, S) \cdot \mathbf{1}[v \notin S]] \leq \frac{1}{k} \Pr_{S \sim \mathcal{D}_{i-1}}[v \notin S], \quad \forall v \in M_i \\
 & \mathbb{E}_{S \sim \mathcal{D}_{i-1}}[x(g_{i,S}^{-1}(v), S) \cdot \mathbf{1}[v \in S]] \geq \frac{1}{k} \Pr_{S \sim \mathcal{D}_{i-1}}[v \in S], \quad \forall v \in \mathcal{N} \\
 & \sum_{v \in M_i} x(v, S) = 1, \quad \forall S \in \text{supp}(\mathcal{D}_{i-1}) \\
 & x(v, S) \geq 0, \quad \forall v \in M_i, S \in \text{supp}(\mathcal{D}_{i-1})
 \end{aligned}$$

- 7: For $x(v, S) > 0$, construct a new distribution:

$$\mathcal{D}_i \leftarrow \{(x(v, S) \cdot \Pr_{\mathcal{D}_{i-1}}[S], S + v - g_{i,S}(v)) \mid v \in M_i, S \in \text{supp}(\mathcal{D}_{i-1})\}.$$

- 8: **return** $\arg\max_{S \in \text{supp}(\mathcal{D}_m)} f(S)$.
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3.2 Modified Greedy Algorithm

In this section, we provide the derandomization Greedy algorithm to solve the meta-submodular maximization problem, as shown in Algorithm 1.

Construct the Matroid

We first add a set D of $2k$ dummy elements to the ground set. That is, for any set $A \in \mathcal{V}$ and $B \in D$ the function satisfies $f(B|A) = 0$. In this way, we can ensure that the set size of each greedy iteration is k . The operation can deal with non-monotone situations, in other words, the target set may contain the elements with negative marginal gain. Finally, the ground set is $\mathcal{N} = \mathcal{V} \cup D$, and the matroid $\mathcal{M} = (\mathcal{N}; \mathcal{I})$ is constructed according to Definition 2. In case of ties between dummy elements and elements from $\mathcal{V}$, always choose the latter.

Algorithm Details

By explicitly maintaining a distribution $\mathcal{D}_i$ in each iteration i , this algorithm will visit the support on the subsets. The key point of Algorithm 1 is that it ensures that the size of the support grows polynomially with i instead of exponentially. Formally, we use a multiset of

164 pairs $\{(P_j, S_j)\}$ to represent distribution $\mathcal{D}_i$, where S_j is a subset and P_j is a probability. In
 165 this representation, we allow $P_j = P_{j'}$, $S_j = S_{j'}$ or both for $j \neq j'$. The sum of P_j equals
 166 to 1. The support of $\mathcal{D}_i$ consists of distinct S_j and is denoted by $\text{supp}(\mathcal{D}_i)$. The number of
 167 pairs in $\mathcal{D}_i$ is denoted by $|\mathcal{D}_i|$. Clearly, $|\text{supp}(\mathcal{D}_i)| \leq |\mathcal{D}_i|$.

168 These constraints characterize the requirements for the elements to be selected in an
 169 iteration. It is easy to see that $x(v, S) = 1/k$ for every $S \in \text{supp}(\mathcal{D}_{i-1})$ and $v \in M_i$ is a
 170 feasible solution of linear program and this is exactly what the random greedy algorithm
 171 does. However, this causes the size of the support of $\mathcal{D}_i$ to grow exponentially with i . By
 172 choosing extreme-point solutions of linear program, Algorithm 1 can ensure that it grows
 173 polynomially rather than exponentially.

174 There are two details about Algorithm 1 to which we need to pay attention. First, for
 175 $A = M_i$ and $B = S$, let $g_{i,S}$ be the mapping defined in Lemma 3. In line 5, Algorithm 1
 176 needs to construct such mappings explicitly. By invoking a perfect matching search algorithm
 177 for bipartite matching, they can be found in polynomial time. Second, in line 7, if $v \in S$,
 178 $S + v - g_{i,S}(v)$ reduces to $S - g_{i,S}(v)$.

179 Linear Constraints

180 Algorithm 1 updates the distribution and constructs $\mathcal{D}_i$ from $\mathcal{D}_{i-1}$ by finding an extreme-point
 181 solution of the linear program. The variables are $x(v, S)$ for $v \in M_i$ and $S \in \text{supp}(\mathcal{D}_{i-1})$,
 182 which can be interpreted as the probability of adding v into S . The constraints can be
 183 divided into six groups:

- 184 ■ The first constraint guarantees that the addition of those v to S with positive $x(v, S)$
 185 attains an average marginal value of elements in M_i . And coefficient $\frac{4^\gamma}{i}$ enables the
 186 meta-submodular functions to search greedily like general submodular functions. Because
 187 of the sensitive submodularity ratio shown in Equation 11.
- 188 ■ The second constraint guarantees that the removal of $g_{i,S}(v)$ from S causes a loss of at
 189 most an average marginal value. It is worth noting that the greedy selection may be
 190 dummy elements in order to satisfy this condition.
- 191 ■ The third group of constraints says that any element outside S will be added into S with
 192 probability at most $1/k$. This constraint ensures that the algorithm runs with polynomial
 193 complexity.
- 194 ■ The fourth group of constraints says that any element in S will be kicked out with
 195 probability at least $1/k$. Elements with negative marginal gain can participate in the
 196 calculation of distribution $\mathcal{D}_i$ without affecting the optimal solution.
- 197 ■ The last two groups ensure that $x(\cdot, S)$ is a legal probability vector for $S \in \text{supp}(\mathcal{D}_{i-1})$.

198 Complexity Analysis

199 Finally, recall that for an extreme point solution x , the number of variables $x(u, S)$ that are
 200 strictly greater than zero is no more than the number of tight constraints. Besides, observe that
 201 the number of constraints at iteration i is at most $n + 3k + 2 + |\text{supp}(\mathcal{D}_{i-1})| \leq n + 3k + 2 + |\mathcal{D}_{i-1}|$.
 202 Since $S + v - g_{i,S}(v)$ is added to $\mathcal{D}_i$ only when $x(u, S) > 0$, the size of $\mathcal{D}_i$ is also no more
 203 than $n + 3k + 2 + |\mathcal{D}_{i-1}|$. On the other hand, in the case of monotone functions, the size of
 204 $\mathcal{D}_i$ can be significantly reduced. Since $|\mathcal{D}_i| = \mathcal{O}(in)$, Algorithm 1 makes $\mathcal{O}(n^2i)$ queries at
 205 iteration i . Therefore, the total number of queries made during all the iterations is $\mathcal{O}(n^2k^2)$.

4 Deterministic Approximation

In this section, we introduce some useful properties that Algorithm 1 can guarantee.

► **Lemma 6.** Assume that $f(\cdot)$ is the meta-submodular function. For any subset $T \in \mathcal{N}$ and a distribution $\mathcal{D}$,

$$\mathbb{E}_{S \sim \mathcal{D}}[f(T \cup S)] \geq f(T) \cdot \min_{v \in \mathcal{N}} \Pr_{S \sim \mathcal{D}}[v \notin S] \quad (12)$$

Proof. Let $\{v_1, v_2, \dots, v_n\}$ denote the elements of $\mathcal{N}$ sorted by a non-decreasing order of the probability $\Pr_{S \sim \mathcal{D}}[v \notin S]$. Additionally, let $A_i = \{v_1, v_2, \dots, v_i\}$ be the set of the first i elements in this order (for every $0 \leq i \leq n$). Then

$$\begin{aligned} \mathbb{E}_{S \sim \mathcal{D}}[f(T \cup S)] &= f(T) + \sum_{i=1}^n \mathbb{E}_{S \sim \mathcal{D}}[1[v_i \in S] \cdot f(v_i \mid T \cup (A_{i-1} \cap S))] \\ &\geq f(T) + \sum_{i=1}^n \mathbb{E}_{S \sim \mathcal{D}}[1[v_i \in S] \cdot f(v_i \mid T \cup A_{i-1})] \\ &= f(T) + \sum_{i=1}^n f(v_i \mid T \cup A_{i-1}) \cdot \Pr_{S \sim \mathcal{D}}[v_i \in S] \\ &= \sum_{i=1}^{n-1} (\Pr_{S \sim \mathcal{D}}[v_i \in S] - \Pr_{S \sim \mathcal{D}}[v_{i+1} \in S]) \cdot f(T \cup A_i) + \\ &\quad \Pr_{S \sim \mathcal{D}}[v_1 \notin S] \cdot f(T) + \Pr_{S \sim \mathcal{D}}[v_n \in S] \cdot f(\mathcal{N}) \\ &\geq \Pr_{S \sim \mathcal{D}}[v_1 \notin S] \cdot f(T) \end{aligned} \quad (13)$$

where the first inequality holds by the submodularity ratio in lemma 5 and the second inequality is based on the fact that $\Pr_{S \sim \mathcal{D}}[v_i \in S]$ is a non-increasing function of i . ◀

► **Lemma 7.** For every iteration $i = 1, 2, \dots, k$ of Algorithm 1, the following properties hold:

- The assignment $x(v, S) = 1/k$ for every $v \in M_i$ and $S \in \text{supp}(\mathcal{D}_{i-1})$ is a feasible solution of linear program.
- The sum of the probabilities in $\mathcal{D}_i$ equals to 1, and therefore $\mathcal{D}_i$ is a valid distribution.
- $|\mathcal{D}_i| \leq n + 3k + 2 + |\mathcal{D}_{i-1}|$. Thus, $|\mathcal{D}_i| \leq ni + 3ki + 2i + 1 = \mathcal{O}(ni)$.

Proof. We prove the lemma by induction on i . Assume that it holds for every $1 \leq i' < i$. First, the correctness of the first property can be verified by directly plugging the assignment into linear program. Next, notice that the sum of the probabilities in $\mathcal{D}_i$ is

$$\sum_{S \in \text{supp}(\mathcal{D}_i)} \Pr_{\mathcal{D}_{i-1}}[S] \sum_{v \in M_i} x(v, S) = \sum_{S \in \text{supp}(\mathcal{D}_i)} \Pr_{\mathcal{D}_{i-1}}[S] = 1 \quad (14)$$

Thus, the second property holds. The third property is concluded in the complexity analysis. ◀

► **Lemma 8.** For every element $v \in \mathcal{N} \setminus D$ and $0 \leq i \leq k$,

$$\Pr_{S \sim \mathcal{D}_i}[v \notin S] \geq \frac{1}{2} \left(1 + \left(1 - \frac{2}{k} \right)^i \right) \quad (15)$$

237 **Proof.** For a fixed $v \in \mathcal{N} \setminus D$,

$$\begin{aligned}
 238 \quad \Pr_{S \sim \mathcal{D}_i} [v \in S] &= \sum_{S \in \text{supp}(\mathcal{D}_{i-1}): v \notin S} \Pr_{\mathcal{D}_{i-1}} [S] \cdot x(v, S) \\
 239 \quad &+ \sum_{S \in \text{supp}(\mathcal{D}_{i-1}): v \in S} \Pr_{\mathcal{D}_{i-1}} [S] \cdot \sum_{v' \in M_i - g_{i,S}^{-1}(v)} x(v', S)
 \end{aligned} \tag{16}$$

240
241 The first part equals to zero if $v \notin M_i$, and if $v \in M_i$, it holds that

$$\begin{aligned}
 242 \quad \sum_{S \in \text{supp}(\mathcal{D}_{i-1}): v \notin S} \Pr_{\mathcal{D}_{i-1}} [S] \cdot x(v, S) &= \sum_{S \in \text{supp}(\mathcal{D}_{i-1})} \Pr_{\mathcal{D}_{i-1}} [S] \cdot x(v, S) \cdot \mathbf{1}[v \notin S] \\
 &= \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [x(v, S) \cdot \mathbf{1}[v \notin S]] \\
 243 \quad &\leq \frac{1}{k} \Pr_{S \sim \mathcal{D}_{i-1}} [v \notin S]
 \end{aligned} \tag{17}$$

244
245 The last inequality holds since x is a feasible solution of linear program.

246 The second part satisfies

$$\begin{aligned}
 247 \quad &\sum_{S \in \text{supp}(\mathcal{D}_{i-1}): v \in S} \Pr_{\mathcal{D}_{i-1}} [S] \cdot \sum_{v' \in M_i - g_{i,S}^{-1}(v)} x(v', S) \\
 &= \sum_{S \in \text{supp}(\mathcal{D}_{i-1}): v \in S} \Pr_{\mathcal{D}_{i-1}} [S] \cdot (1 - x(g_{i,S}^{-1}(v), S)) \\
 &= \Pr_{S \sim \mathcal{D}_{i-1}} [v \in S] - \sum_{S \in \text{supp}(\mathcal{D}_{i-1})} \Pr_{\mathcal{D}_{i-1}} [S] \cdot x(g_{i,S}^{-1}(v), S) \cdot \mathbf{1}[v \in S] \\
 &= \Pr_{S \sim \mathcal{D}_{i-1}} [v \in S] - \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [x(g_{i,S}^{-1}(v), S) \cdot \mathbf{1}[v \in S]] \\
 248 \quad &\leq (1 - 1/k) \Pr_{S \sim \mathcal{D}_{i-1}} [v \in S]
 \end{aligned} \tag{18}$$

249
250 The last inequality holds since x is a feasible solution of linear program.

251 Let $P_{i,v} = \Pr_{S \sim \mathcal{D}_i} [v \in S]$. By the above argument,

$$252 \quad P_{i,v} \leq (1 - P_{i-1,v})/k + P_{i-1,v}(1 - 1/k) = P_{i-1,v}(1 - 2/k) + 1/k \tag{19}$$

254 For $v \in \mathcal{N} \setminus D$, $P_{0,v} = 0$. It is easy to show by induction that

$$255 \quad P_{i,v} \leq 0.5 \cdot (1 - (1 - 2/k)^i) \tag{20}$$

256
257 The lemma follows immediately. $\blacktriangleleft$

258 Let O be the optimal solution. We have

259 **► Lemma 9.** For every iteration $i = 1, 2, \dots, k$ of Algorithm 1,

$$260 \quad \mathbb{E}_{S \sim \mathcal{D}_i} [f(S)] \geq \left(1 - \frac{2}{k}\right) \cdot \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [f(S)] + \frac{(1 + (1 - 2/k)^{i-1})}{2k} \cdot f(O) \tag{21}$$

262
263 **Proof.** On the one hand,

$$\begin{aligned}
 264 \quad \sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [x(v, S) \cdot f(v \mid S)] &\geq \frac{1}{k} \sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} \left[\frac{4^\gamma}{i} \cdot f(v \mid S) \right] \\
 &\geq \frac{1}{k} \sum_{v \in O} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} \left[\frac{4^\gamma}{i} \cdot f(v \mid S) \right] \\
 &\geq \frac{1}{k} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [f(O \cup S) - f(S)]
 \end{aligned}$$

$$\geq \frac{1}{k} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [0.5 \cdot (1 + (1 - 2/k)^{i-1}) \cdot f(O) - f(S)] \quad (22)$$

The first inequality holds since x is a feasible solution of linear program. The second is due to the choice of M_i . The third is due to the sensitive submodularity ratio in Equation 11.

The last follows from Lemma 6 and Lemma 8.

On the other hand,

$$\begin{aligned} & \sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [x(v, S) \cdot f(g_{i,S}(v) \mid S \setminus \{g_{i,S}(v)\})] \\ & \leq \frac{1}{k} \sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} \left[\frac{4^\gamma}{i} \cdot f(g_{i,S}(v) \mid S \setminus \{g_{i,S}(v)\}) \right] \\ & = \frac{1}{k} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} \left[\sum_{v \in M_i} \frac{4^\gamma}{i} \cdot (f(S) - f(S \setminus \{g_{i,S}(v)\})) \right] \\ & \leq \frac{1}{k} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [f(S)] \end{aligned} \quad (23)$$

The first inequality holds since x is a feasible solution of linear program. The last inequality is due to the sensitive submodularity ratio in Equation 11.

Finally, by combining the above inequalities,

$$\begin{aligned} & \mathbb{E}_{S \sim \mathcal{D}_i} [f(S)] \\ & = \sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [x(v, S) \cdot f(S + v - g_{i,S}(v))] \\ & \geq \sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [x(v, S) \cdot (f(S + v) + f(S - g_{i,S}(v)) - f(S))] \\ & = \sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [x(v, S) \cdot (f(S) + f(v \mid S) - f(g_{i,S}(v) \mid S \setminus \{g_{i,S}(v)\}))] \\ & = \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [f(S)] + \sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [x(v, S) \cdot f(v \mid S)] \\ & \quad - \sum_{v \in M_i} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [x(v, S) \cdot f(g_{i,S}(v) \mid S \setminus \{g_{i,S}(v)\})] \\ & \geq \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [f(S)] + \frac{1}{k} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [0.5 \cdot (1 + (1 - 2/k)^{i-1}) \cdot f(O) - f(S)] \\ & \quad - \frac{1}{k} \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [f(S)] \\ & = \left(1 - \frac{2}{k}\right) \cdot \mathbb{E}_{S \sim \mathcal{D}_{i-1}} [f(S)] + \frac{(1 + (1 - 2/k)^{i-1})}{2k} \cdot f(O) \end{aligned} \quad (24)$$

The first inequality is according to the definition of meta-submodular functions. $\blacktriangleleft$

The following theorem provides a theoretical guarantee for Algorithm 1.

► **Theorem 10.** *Algorithm 1 achieves a $(1 + e^{-2})/4 - \mathcal{O}(1/k^2)$ approximation ratio using $\mathcal{O}(n^2 k^2)$ queries.*

Proof. We prove by induction that

$$\mathbb{E}[f(S_i)] \geq \frac{1}{4} \left[1 + \left(\frac{2(i+1)}{k} - 1 \right) \left(1 - \frac{2}{k} \right)^{i-1} \right] \cdot f(O) \quad (25)$$

Clearly, the claim holds for $i = 0$. Assume that the claim holds for every $0 \leq i' < i$. By the above lemma,

$$\begin{aligned}
 \mathbb{E}_{S \sim \mathcal{D}_i}[f(S)] &\geq \left(1 - \frac{2}{k}\right) \cdot \mathbb{E}_{S \sim \mathcal{D}_{i-1}}[f(S)] + \frac{0.5(1 + (1 - 2/k)^{i-1})}{k} \cdot f(O) \\
 &\geq \left(1 - \frac{2}{k}\right) \cdot \frac{1}{4} \left[1 + \left(\frac{2i}{k} - 1\right) \left(1 - \frac{2}{k}\right)^{i-2}\right] \cdot f(O) \\
 &\quad + \frac{1 + (1 - 2/k)^{i-1}}{2k} \cdot f(O) \\
 &= \frac{1}{4} \left[1 + \left(\frac{2(i+1)}{k} - 1\right) \left(1 - \frac{2}{k}\right)^{i-1}\right] \cdot f(O)
 \end{aligned} \tag{26}$$

Therefore,

$$\begin{aligned}
 \mathbb{E}_{S \sim \mathcal{D}_k}[f(S)] &\geq \frac{1}{4} \left[1 + \left(\frac{2(k+1)}{k} - 1\right) \left(1 - \frac{2}{k}\right)^{k-1}\right] \cdot f(O) \\
 &\geq \frac{1 + \frac{e^{-2}(1+2/k)(1-4/k)}{1-2/k}}{4} \cdot f(O) \\
 &= \left(\frac{1 + e^{-2}}{4} - \mathcal{O}(1/k^2)\right) \cdot f(O)
 \end{aligned} \tag{27}$$

The second inequality is according to the $(1 - \frac{1}{x})^x \leq e^{-1}$. In short, Theorem 1 and Theorem 10 are proved with a constant bound when k is very large. $\blacktriangleleft$

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